BJA

British Journal of Anaesthesia, 134 (2): 281–287 (2025)

doi: 10.1016/j.bja.2024.10.007
Advance Access Publication Date: 17 December 2024
Editorial

EDITORIAL

Improving lung protective mechanical ventilation: the individualised intraoperative open-lung approach

Carlos Ferrando[1,2,*], Jordi Vallverdú[3], Luigi Zattera[3], Gerardo Tusman[4] and Fernando Suárez-Sipmann[2,5]

[1]Department of Anaesthesia and Critical Care, Hospital Clínic, Institut d'Investigació August Pi i Sunyer, Barcelona, Spain, [2]CIBER de Enfermedades Respiratorias, Instituto de Salud Carlos III, Madrid, Spain, [3]Department of Anaesthesia and Critical Care, Hospital Clínic, Barcelona, Spain, [4]Department of Anaesthesia, Hospital Privado de Comunidad, Mar del Plata, Argentina and [5]Intensive Care Unit, Hospital Universitario La Princesa, Madrid, Spain

*Corresponding author. E-mail: cafeoranestesia@gmail.com

Summary

Despite the maturity and sophistication of anaesthesia workstations, improvements in our understanding of intraoperative mechanical ventilation, and use of less invasive surgical techniques, postoperative pulmonary complications (PPCs) are still a common problem in surgical patients of all ages. PPCs are associated with a higher incidence of perioperative morbidity and mortality, longer hospital stays, and higher healthcare costs. PPCs are strongly associated with anaesthesia-induced atelectasis, which predisposes to lung damage when partially collapsed lungs are subjected to mechanical ventilation. Lung protective ventilation is thus a modifiable factor that can positively impact the incidence of PPCs after surgery. Intraoperative protective ventilation strategies have been based on two main but intrinsically different hypotheses: one based on sole reduction of tidal volume and pressures, using minimal positive end-expiratory pressure (PEEP), tolerating the presence of lung collapse, and the other also limiting tidal volume and pressures after actively resolving atelectasis by lung recruitment and PEEP individualisation, the individualised open-lung approach. We review the concepts of the individualised open-lung approach, its potential benefits, and outstanding questions. We conclude with a proposal for personalised lung protective ventilation.

Keywords: alveolar recruitment manoeuvre; atelectasis; lung protective ventilation; open-lung; positive end-expiratory pressure; postoperative pulmonary complications

Postoperative pulmonary complications (PPCs) include a spectrum of clinical manifestations that range from mild postoperative hypoxaemia to severe acute respiratory failure, all of which require postoperative respiratory support, from supplemental oxygen to tracheal intubation and invasive mechanical ventilation.[1] PPCs constitute an important public health problem given the many inciting interventions performed daily and the high prevalence of these complications. They occur in 5% of patients undergoing general anaesthesia and in up to 25% of high-risk patients undergoing major surgery, in both cases defined by various risk scales.[2,3] They are also associated with an increase in unplanned admission to the intensive care unit (ICU), prolonged hospital stay, and mortality.[2] Estimates suggest that development of PPCs can increase perioperative mortality up to five-fold.[3]

Various interventions on modifiable risk factors have the potential to reduce the risk of PPCs.[2,3] Among all these modifiable risk factors, lung protective mechanical ventilation is probably one of the most important because of its direct effect on the lungs and its potential for improvement. As the most prevalent postoperative PPC is atelectasis, which can also be a factor in development of other PPCs,[4,5] establishing an open-lung strategy by applying recruitment manoeuvres and individualised positive end-expiratory pressure (iPEEP) should reduce the risk for PPCs. Based on this hypothesis, and on the results of the iPROVE study in patients undergoing abdominal surgery,[6] we proposed a clinical trial to demonstrate the benefits of an individualised open-lung approach (iOLA) in patients undergoing lung resection surgery with one-lung ventilation.[7] In that study, we showed that, compared with a standardised lung protection strategy, the iOLA reduced PPCs. This is in line with the growing evidence supporting the use of personalised lung protective mechanical ventilation based on recruitment manoeuvres and iPEEP. From a physiological point of view, all the studies in which iPEEP has been compared with a pragmatic fixed PEEP or zero PEEP in surgical patients have consistently demonstrated improvement in respiratory system and lung mechanics, lung volumes, oxygenation, and lung efficiency.[8–14] From a clinical perspective, in several studies the iOLA has effectively reduced the incidence of PPCs in moderate-to-high risk patients.[6,7,15–17] However, solid evidence is still lacking in a number of surgical procedures. In addition, as various studies have shown, performing recruitment manoeuvres and applying iPEEP is no guarantee of reducing PPCs.[18,19]

The individualised open-lung approach

The individualised open-lung approach (iOLA) is a lung protective mechanical ventilation strategy aimed at minimising, or ideally eliminating, lung collapse.[20] This approach consists of two clearly distinct sequential phases: (1) a lung opening phase by means of alveolar recruitment manoeuvres to open collapsed alveoli; followed by (2) iPEEP to maintain lung expansion avoiding de-recruitment.[20] Personalising the PEEP can also minimise the risk of high PEEP-induced alveolar overdistension by restoring a normal functional lung volume.[20]

Alveolar recruitment manoeuvre

The recruitment manoeuvre is an adjuvant ventilatory treatment that consists of a brief, transient, controlled increase in transpulmonary pressure aimed at re-expanding collapsed lung areas. The basic principle of recruitment manoeuvres is to apply a sufficient high inspiratory transpulmonary pressure (Ptp[insp]), called opening pressure, for a sufficiently effective period of time. As Ptp(insp) is not routinely monitored, airway pressure (Paw) is used as a surrogate, as an increase in Paw generates a proportional increase in Ptp(insp). The physics around lung collapse and recruitment conforms to Laplace's Law, which explains why a collapsed unit needs high internal pressure to open but a lower pressure to remain open afterward.[21] Thus, it is a dynamic process based on overcoming a pressure threshold (i.e. the opening pressure that achieves an immediate physiological, structural, and functional change in the lung). In healthy lungs, Paw of 40 cm H_2O is sufficient to open collapsed alveoli. However, in certain circumstances it might be necessary to apply higher airway pressures to achieve effective transpulmonary opening pressure, such as for patients with obesity or laparoscopic surgery.[8] One of the most difficult and controversial calculations to be made when planning a clinical recruitment manoeuvre is to determine the level and duration of the pressure required to achieve an effective and safe intervention, hence the wide variety of recruitment manoeuvre approaches described in the literature.

How should recruitment manoeuvres be performed?

Although many recruitment manoeuvres have been described,[22] the most widely accepted is the step-wise (incremental) PEEP manoeuvre in a pressure-controlled ventilation (PCV) mode, in which the normal inspiratory-expiratory cycle is preserved (Fig. 1a). The ventilation mode is changed to PCV (if it was not previously set), and settings are changed to a fixed driving pressure of 20 cm H_2O, a ventilatory frequency of between 10 and 20 cycles min^{-1}, and usually an inspiratory to expiratory (I:E) ratio of 1:1. The gradual increase in pressure begins by increasing PEEP from a baseline of 5 cm H_2O (or higher if previously adjusted) to 10, 15, and then 20 cm H_2O (or higher values in special conditions such as obesity, Trendelenburg position or robotic surgery), maintaining each level for three to five respiratory cycles. The resulting opening pressure of 40 cm H_2O is maintained for 30 s.

How is the effectiveness of a recruitment manoeuvre evaluated?

The immediate consequences of an effective recruitment manoeuvre are an increased end-expiratory lung volume (EELV), improved lung compliance, and improved oxygenation as a result of an increase in the gas exchange surface area. From a clinical point of view, the classical criteria used to define effectiveness are based on changes in oxygenation.[23] For example, an open lung has been defined as a Pa_{O_2}/FiO_2 ratio of >400 when 1.0 FiO_2, or $SpO_2 \geq 97\%$ when FiO_2 0.21, equivalent to a shunt of <10%.[24] In healthy lungs during anaesthesia, shunt is closely related to the amount of atelectasis,[25] so a reduction of shunt is a good indicator of recovery of EELV. This, together with improvement in lung compliance, creates a much better lung mechanical scenario to implement lung protective ventilation.

Individualised positive end-expiratory pressure

The respiratory system reaches its resting state at end-expiration, when respiratory muscle activity and air movement cease and the pressure of the entire patent airway equals atmospheric pressure. The volume remaining in the lungs during this resting state is the functional residual capacity (FRC), or EELV with mechanical ventilation, as PEEP changes this equilibrium by increasing the 'working' pressure of the respiratory system to a pre-determined value that exceeds atmospheric pressure. As a result, intrathoracic, alveolar, and transpulmonary pressure increase during the respiratory cycle. End-expiratory alveolar pressure (PEEP) maintains the alveolar units participating in gas exchange open, overcoming their tendency to collapse during mechanical ventilation.[22]

However, PEEP titration continues to be one of the most controversial aspects of mechanical ventilation, as its known beneficial effects have to be balanced against its potential negative effects. Inappropriately high or low PEEP can impair haemodynamics, compromising cardiac output, and can increase the risk of ventilator-induced lung injury.[22] Excessive PEEP can result in alveolar overdistension, and insufficient PEEP failure to prevent lung collapse.[22] Numerous studies have analysed the level that results in the best compromise between these beneficial and negative effects. One of the methods that has recently gained the most attention is iPEEP. This approach is particularly attractive in the context of lung

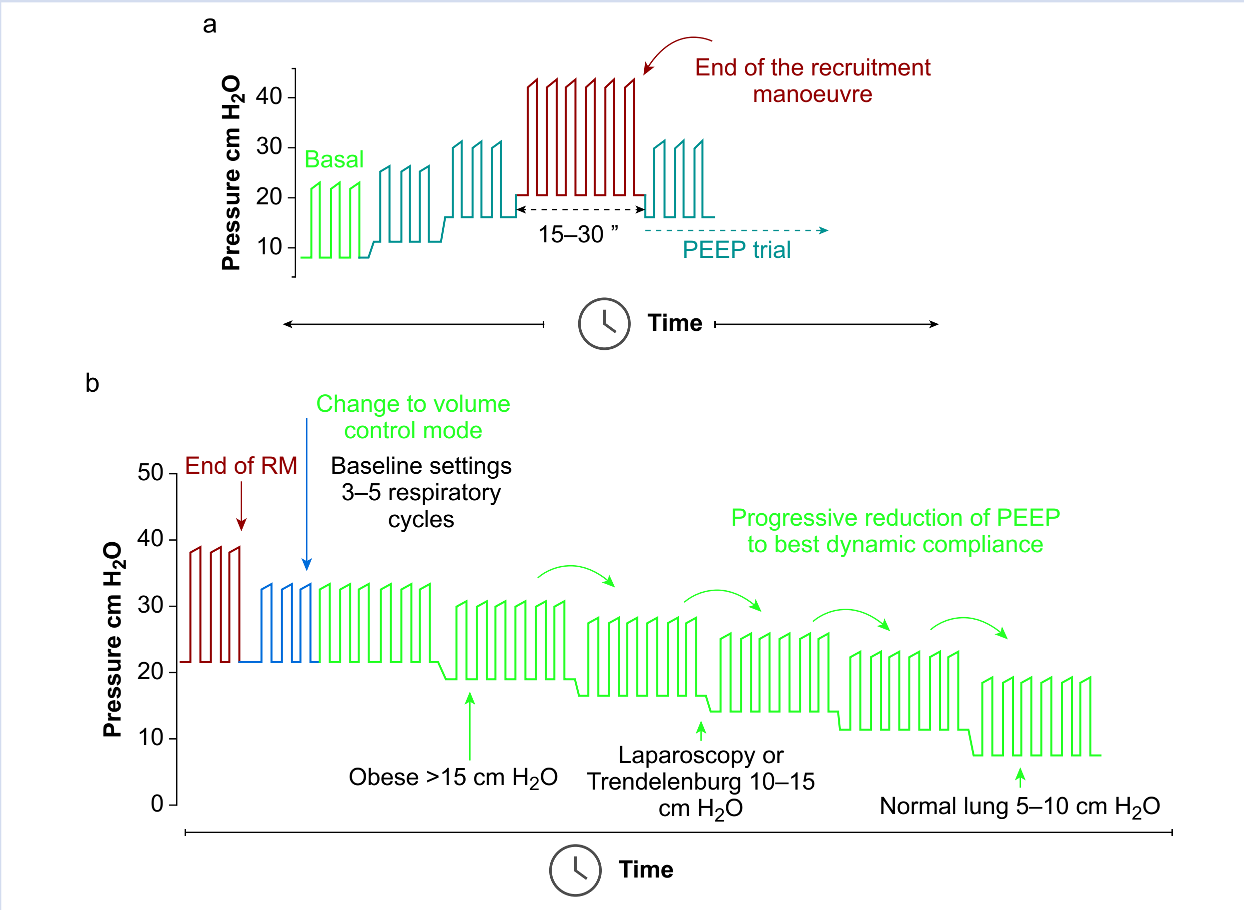

Fig 1. Recruitment manoeuvre and PEEP titration trial. (a) Stepwise pressure-controlled ventilation (PCV) cycling manoeuvre for healthy lungs. From the baseline situation, ventilation mode is changed to PCV (if it was not previously set). Settings are changed to fixed driving pressure of 20 cm H_2O, a ventilatory frequency of between 10 and 20 cycles min^{-1} and, in general, an inspiratory to expiratory ratio (I:E) of 1:1. The gradual increase in pressure begins by increasing PEEP from the baseline level (5 cm H_2O) to 10, 15 and until reaching a value of 20 cm H_2O (evaluate higher values in special conditions such as obese patients, Trendelenburg position or robotic surgery, among others), maintaining each level for three to five respiratory cycles. The resulting opening pressure of 40 cm H_2O (marked in red) is maintained for 30 s. (b) PEEP titration trial. After completing the RM, the first step is to immediately reduce both the inspiratory pressure and PEEP level (respiratory cycles in blue in Fig. 1b). At the start of the trial, PEEP is set to a level appropriate to the patient's pulmonary status or clinical characteristics. In healthy patients, the test can be started at a PEEP level of 16 cm H_2O. In specific circumstances, such as laparoscopy, Trendelenburg positioning, or in obese patients (BMI $\geq$35) it is recommended to start at 20 cm H_2O. After a brief stabilisation period, the ventilatory mode should ideally be changed to volume-controlled ventilation (cycles in blue), although this is not strictly necessary, as individualised PEEP (iPEEP) can also be titrated in a pressure-control mode. The remaining ventilatory parameters should be set to mimic as far as possible the parameters to be used during ventilation. Once the parameters have been selected and set, the PEEP level is gradually reduced in decrements of 2 cm H_2O while the other parameters remain unchanged. Each level is maintained for around 15–30 s. Cdyn (alternatives that can be used are: driving pressure, SpO_2, lung ultrasound), must be monitored continuously during titration to determine the point of maximum compliance, which corresponds to the point of collapse of the most dependent lung regions. This is the iPEEP level. PEEP, positive end-expiratory pressure; RM, recruitment manoeuvre.

protective ventilation, as it allows the anaesthesiologist to apply iPEEP that minimises alveolar collapse. The ideal individualised setting according to the best lung compliance should result in a good balance between overdistension and collapse and result in the lowest driving pressure for a given tidal volume, which in turn will reduce the risk of PPCs.[26]

Which method should be used to individualised PEEP?

The decremental PEEP titration method has been proposed to individualise PEEP. Performed after a recruitment manoeuvre (Fig. 1b), it aims to determine the lung closing pressure immediately (i.e. the PEEP level at which the recruited lung, the most unstable recruited alveolar units, start to collapse). During titration the ventilation mode is generally changed to volume-controlled ventilation (cycles in blue), although this is not strictly necessary, as iPEEP can also be titrated in pressure control mode. The remaining ventilatory parameters should be set as similar as possible to the parameters set for ventilation.[6,7] Once the parameters have been adjusted, the initial PEEP level is selected according to the patient's pulmonary status and other patient-specific factors or clinical characteristics such as body mass index. In healthy patients, an initial PEEP of 16 cm H_2O is enough.[27] In specific circumstances, such as laparoscopy, Trendelenburg positioning, or obesity, it is recommended to start at 18–20 cm H_2O. Then, PEEP is sequentially reduced generally in 2 cm H_2O steps for 15–30 s while maintaining all other ventilation parameters unchanged until the lung's closing pressure is detected.[6,7,28]

How is iPEEP determined?

Several methods can be used to detect lung closing pressure. Respiratory mechanics (respiratory system compliance, driving pressure, and transpulmonary pressure), oxygenation (Pa_{O_2} and SpO_2), lung efficiency (volumetric capnography), and imaging techniques (lung ultrasound and electrical impedance tomography) have been described.[28,29] In surgical patients, a multimodal noninvasive approach of continuous variables can be used to individualise PEEP.[28] The parameters most widely used are those related to respiratory mechanics, such as dynamic lung compliance (Cdyn). Cdyn has the advantage of being available in any ventilator and is monitored continuously during the PEEP titration trial, allowing to identify the point of maximum compliance, which corresponds to the closing pressure (Fig. 1b). Another interesting method for normal lungs during anaesthesia is the use of SpO_2. Low FiO_2 (0.21) can be used to detect the presence or absence of shunt related to atelectasis with an SpO_2 value of <97% indicating a shunt >10%.[24] In morbidly obese patients, SpO_2 was more accurate than respiratory system compliance (Crs) and EELV in detecting recruitment and lung collapse in anaesthetised patients.[8,28]

Individualised open-lung approach: unanswered questions for precision medicine

Although several physiological studies and clinical trials have broadened our understanding of personalised protective mechanical ventilation based on an iOLA strategy in surgical patients, there is still a long way to go. We believe there remain a number of important open questions that should be answered in future clinical trials.

Should alveolar recruitment manoeuvres be performed in all surgery patients?

In most large studies, the physiological rationale for performing recruitment manoeuvres, with or without subsequent iPEEP titration, is that up to 90% of patients undergoing general anaesthesia and mechanical ventilation develop intraoperative atelectasis.[30] However, in a secondary analysis of the iPROVE and iPROVE-O2 studies, 33% of the 1480 included patients did not develop significant atelectasis as defined by Pa_{O_2}/FiO_2 >400 with low PEEP ventilation and without recruitment manoeuvres.[31] This study involved abdominal surgery patients, a population particularly susceptible to developing intraoperative atelectasis. Another physiological study found no differences in oxygenation, ventilatory efficiency, or respiratory mechanics between patients who initially presented open-lung conditions and those who did so after undergoing recruitment manoeuvres and iPEEP titration.[8] This suggests that a large proportion of conventional surgery patients do not develop significant lung atelectasis and would thus not benefit from recruitment manoeuvres.

Which patients should undergo alveolar recruitment manoeuvres?

Recruitment manoeuvres should only be performed on patients presenting with intraoperative atelectasis. Therefore, the first step should always be to diagnose the presence or absence of atelectasis. Several methods can be used to diagnose intraoperative atelectasis by either imaging (lung ultrasound, electrical impedance tomography), arterial blood gas measurement (Pa_{O_2}/FiO_2), or the air test (SpO_2). An SpO_2 ≤96% with FiO_2 to 0.21 in an otherwise healthy lung indicates the presence of collapse-related shunt >10%.[24] This noninvasive method has been validated in both adults and paediatric patients.[25,32]

Should opening pressure during recruitment manoeuvres be individualised?

Recruitment manoeuvres are performed using different opening pressures for a specific time. In a healthy lung, a Paw of 40 cm H_2O is usually recommended. However, alveolar opening pressures can differ considerably in specific surgical conditions or patients. Studies performed in patients undergoing laparoscopic abdominal surgery or in patients with obesity confirmed mean opening pressure values of 40 and 45 cm H_2O, respectively.[8,33] Failing to reach the alveolar opening pressure during the recruitment manoeuvre will have no physiological benefits and thus will likely not impact the incidence of PPCs. It follows that the opening pressure should be individualised ideally. A point of care use of the air test could be a simple and optimal method of defining the alveolar opening pressure in haemodynamically stable patients.[8] However, this approach warrants further validation in more types of surgery and patient populations, and clinicians should understand and be aware of its limitations.

Should the open-lung condition be re-evaluated during surgery?

The effectiveness and intraoperative maintenance of an iOLA strategy should be frequently re-assessed both after performing the recruitment manoeuvre and individualising PEEP. This is because of the dynamic conditions encountered during surgery, especially with the introduction of more advanced techniques such as robotic surgery. Various studies have shown that a high number of patients require repeat recruitment manoeuvres, iPEEP titration, or both during the intervention to maintain an open-lung condition.[6,7]

Personalised intraoperative open-lung approach algorithm

A proposal for a personalised ventilatory management is shown in Figure 2. It involves a sequential approach consisting of four steps. Step 1: evaluate lung conditions. Apart from lung compliance and oxygenation, assessment of room air SpO_2 (≤96%) and lung ultrasound (dependent juxtapleural consolidation pattern and absence of lung sliding) can reliably diagnose the presence of perioperative atelectasis. Step 2: ensure haemodynamic stability before the recruitment manoeuvre, as this is a reason for aborting further manoeuvres. Step 3: implement the iOLA strategy starting with an individualised recruitment manoeuvre by a stepwise increase in Paw to restore functional lung volume. Opening pressure could be individualised by using SpO_2 measured on room air, with SpO_2 >96% indicating a shunt <10%. After the individualised

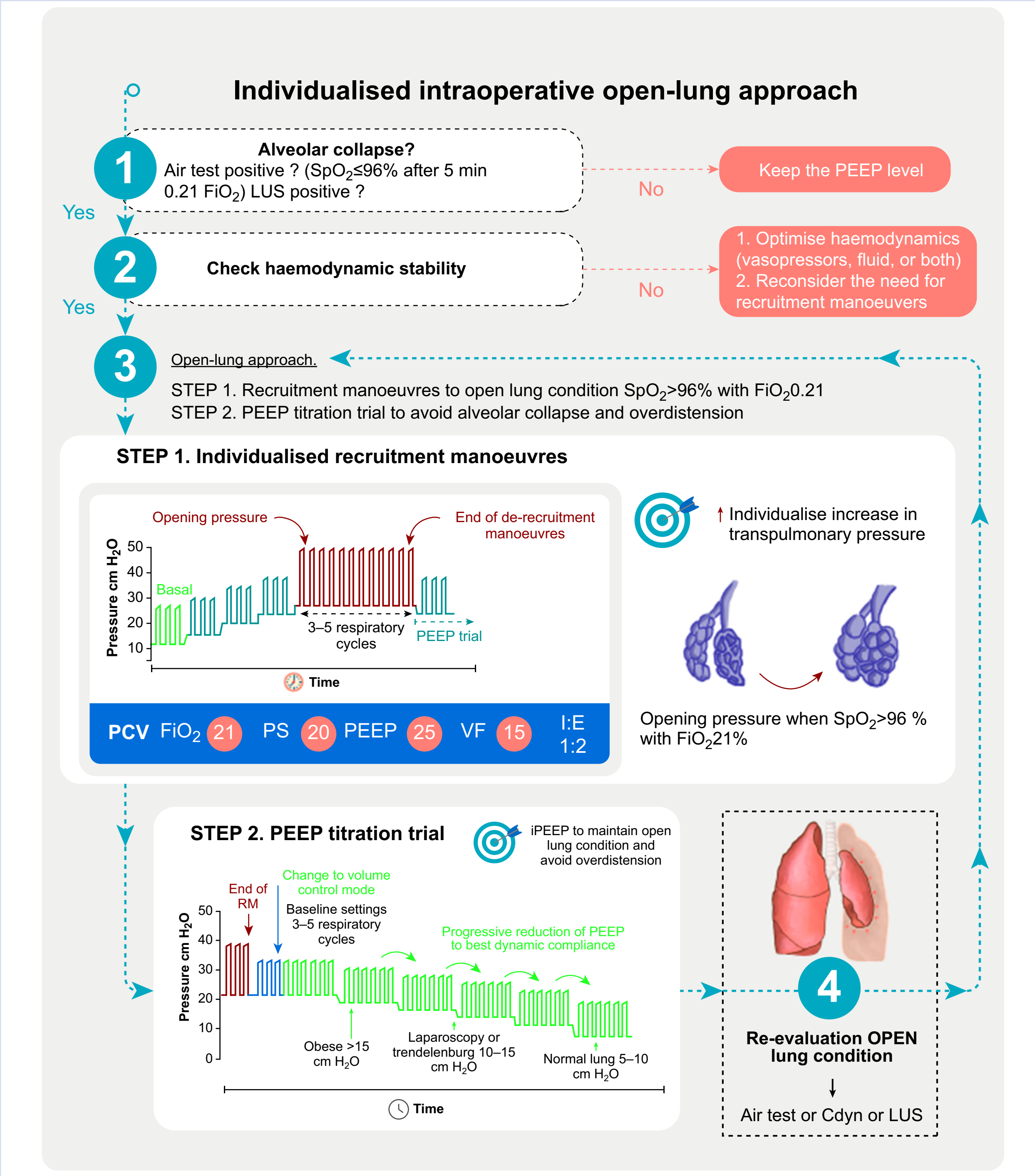

Fig 2. Individualised intraoperative open-lung approach algorithm. Cdyn, respiratory system dynamic compliance; FRC, functional residual capacity; FiO_2, inspired oxygen fraction; I:E, inspiratory to expiratory relationship; LUS, lung ultrasound; PCV, pressure-controlled ventilation; PEEP, positive end-expiratory pressure; PS, pressure support; RM, recruitment manoeuvre; SpO_2, peripheral oxyhaemoglobin saturation; VF, ventilatory frequency.

recruitment manoeuvres, select iPEEP through the PEEP titration trial (Fig. 1b). Step 4: evaluate lung condition during surgery and the need for new individualised recruitment manoeuvre. A positive air test (SpO_2 <97% on room air), a decrease in Cdyn >10%, or both suggest alveolar re-collapse.

Conclusions

There is growing evidence that personalised lung protective mechanical ventilation based on recruitment manoeuvres and individualised PEEP reduces the incidence of postoperative pulmonary complications. However, evidence to generalise this strategy to all patients undergoing surgery is still lacking. Further research is needed to better personalise mechanical ventilation to the singular conditions of each patient and each type of surgery.

Acknowledgements

We thank Adriana Jacas (JACAS@clinic.cat) from the Anaesthesiology and Critical Care Department of the Hospital Clinic of Barcelona for her invaluable help in the design and creation of the figures.

Declaration of interest

The authors declare that they have no conflict of interest.

References

1. Abbott TEF, Fowler AJ, Pelosi P, et al. A systematic review and consensus definitions for standardised end-points in perioperative medicine: pulmonary complications. *Br J Anaesth* 2018; **120**: 1066–79
2. Canet J, Gallart L, Gomar C, et al. Prediction of postoperative pulmonary complications in a population-based surgical cohort. *Anesthesiology* 2010; **113**: 1338–50
3. Neto AS, Da Costa LGV, Hemmes SNT, et al. The LAS VEGAS risk score for prediction of postoperative pulmonary complications: an observational study. *Eur J Anaesthesiol* 2018; **35**: 691–701
4. Ferrando C, Suárez-Sipmann F, Librero J, et al. A noninvasive postoperative clinical score to identify patients at risk for postoperative pulmonary complications: the air-test score. *Minerva Anestesiol* 2020; **86**: 404–15
5. Miskovic A, Lumb AB. Postoperative pulmonary complications. *Br J Anaesth* 2017; **118**: 317–34
6. Ferrando C, Soro M, Unzueta C, et al. Individualised perioperative open-lung approach versus standard protective ventilation in abdominal surgery (iPROVE): a randomised controlled trial. *Lancet Respir Med* 2018; **6**: 193–203
7. Ferrando C, Carramiñana A, Piñeiro P, et al. Individualised, perioperative open-lung ventilation strategy during one-lung ventilation (iPROVE-OLV): a multicentre, randomised, controlled clinical trial. *Lancet Respir Med* 2024; **12**: 195–206
8. Ferrando C, Tusman G, Suarez-Sipmann F, et al. Individualized lung recruitment maneuver guided by pulse-oximetry in anesthetized patients undergoing laparoscopy: a feasibility study. *Acta Anaesthesiol Scand* 2018; **62**: 608–19
9. Fernandez-Bustamante A, Sprung J, Parker RA, et al. Individualized PEEP to optimise respiratory mechanics during abdominal surgery: a pilot randomised controlled trial. *Br J Anaesth* 2020; **125**: 383–92
10. Pereira SM, Tucci MR, Morais CCA, et al. Individual positive end-expiratory pressure settings optimize intraoperative mechanical ventilation and reduce postoperative atelectasis. *Anesthesiology* 2018; **129**: 1070–81
11. Van Hecke D, Bidgoli JS, Van der Linden P. Does lung compliance optimization through PEEP manipulations reduce the incidence of postoperative hypoxemia in laparoscopic bariatric surgery? A randomized trial. *Obes Surg* 2019; **29**: 1268–75
12. Ferrando C, Mugarra A, Gutierrez A, et al. Setting individualized positive end-expiratory pressure level with a positive end-expiratory pressure decrement trial after a recruitment maneuver improves oxygenation and lung mechanics during one-lung ventilation. *Anesth Analg* 2014; **118**: 657–65
13. Ferrando C, Suarez-Sipmann F, Tusman G, et al. Open lung approach versus standard protective strategies: effects on driving pressure and ventilatory efficiency during anesthesia - a pilot, randomized controlled trial. *PLoS One* 2017; **12**, e0177399
14. Nestler C, Simon P, Petroff D, et al. Individualized positive end-expiratory pressure in obese patients during general anaesthesia: a randomized controlled clinical trial using electrical impedance tomography. *Br J Anaesth* 2017; **119**: 1194–205
15. Mini G, Ray BR, Anand RK, et al. Effect of driving pressure-guided positive end-expiratory pressure (PEEP) titration on postoperative lung atelectasis in adult patients undergoing elective major abdominal surgery: a randomized controlled trial. *Surgery* 2021; **170**: 277–83
16. Zhang C, Xu F, Li W, et al. Driving pressure-guided individualized positive end-expiratory pressure in abdominal surgery: a randomized controlled trial. *Anesth Analg* 2021; **133**: 1197–205
17. Yu J, Wu Z, An R, et al. Association between driving pressure and postoperative pulmonary complications in patients undergoing lung resection surgery: a randomised clinical trial. *Anaesth Crit Care Pain Med* 2023; **42**, 101160
18. Park M, Yoon S, Nam JS, et al. Driving pressure-guided ventilation and postoperative pulmonary complications in thoracic surgery: a multicentre randomised clinical trial. *Br J Anaesth* 2023; **130**: e106–18
19. Ernest EE, Bhattacharjee S, Baidya DK, et al. Effect of incremental PEEP titration on postoperative pulmonary complications in patients undergoing emergency laparotomy: a randomized controlled trial. *J Clin Monit Comput* 2024; **38**: 445–54
20. Hubmayr RD, Kallet RH. Understanding pulmonary stress-strain relationships in severe ARDS and its implications for designing a safer approach to setting the ventilator. *Respir Care* 2018; **63**: 219–26
21. Lachmann B. Open the lung and keep the lung open. *Intensive Care Med* 1992; **18**: 319
22. Hess D. Recruitment maneuvers and PEEP titration. *Respir Care* 2015; **60**: 1688–704
23. Reske A, Costa E, Reske A, et al. Bedside estimation of nonaerated lung tissue using blood gas analysis. *Crit Care Med* 2013; **41**: 732–43
24. Jones JG, Jones SE. Discriminating between the effect of shunt and reduced VA/Q on arterial oxygenation is particularly useful in clinical practice. *J Clin Monit Comput* 2000; **16**: 337–50
25. Ferrando C, Romero C, Tusman G, et al. The accuracy of postoperative, non-invasive Air-Test to diagnose atelectasis in healthy patients after surgery: a prospective, diagnostic pilot study. *BMJ Open* 2017; **7**, e015560
26. Serpa Neto A, Hemmes S, Barbas C, et al. Association between driving pressure and development of postoperative pulmonary complications in patients undergoing mechanical ventilation for general anaesthesia: a meta-analysis of individual patient data. *Lancet Respir Med* 2016; **4**: 272–80

27. Laguna G, Suárez-Sipmann F, Tusman G, et al. Rationale and study design for an Individualized PeriopeRative Open lung VEntilatory approach in Emergency Abdominal Laparotomy/scopy: study protocol for a prospective international randomized controlled trial. *Rev Esp Anestesiol Reanim (Engl Ed)* 2024; **71**: 445–53
28. Tusman G, Acosta C, Ochoa M, et al. Multimodal non-invasive monitoring to apply an open lung approach strategy in morbidly obese patients during bariatric surgery. *J Clin Monit Comput* 2020; **34**: 1015–24
29. Esposito T, Fregonese M, Morettini G, et al. Intraoperative individualization of positive-end-expiratory. Pressure through electrical impedance tomography or esophageal pressure assessment: a systematic review and meta-analysis of randomized controlled trials. *J Clin Monit Comput* 2024; **38**: 89–100
30. Lundquist H, Hedenstierna G, Strandberg A, Tokics L, Brismar B. CT-assessment of dependent lung densities in man during general anaesthesia. *Acta Radiol* 1995; **36**: 626–32
31. Ferrando C, Librero J, Tusman G, et al. Intraoperative open lung condition and postoperative pulmonary complications. A secondary analysis of iPROVE and iPROVE-O2 trials. *Acta Anaesthesiol Scan* 2022; **66**: 30–9
32. González-Pizarro P, Acosta C, Alcaraz G, et al. Clinical validation of the Air Test for the non-invasive detection of perioperative atelectasis in children. *Minerva Anestesiol* 2024; **90**: 635–43
33. Tusman G, Groisman I, Fiolo FE, et al. Noninvasive monitoring of lung recruitment maneuvers in morbidly obese patients: the role of pulse oximetry and volumetric capnography. *Anesth Analg* 2014; **118**: 137–44